

APOORV DIXIT

apoorvdixit619@gmail.com | Jersey City, NJ | +1 (213) 574-5136 | linkedin.com/in/apoorvdixit9 | github.com/apoorvdixit99

EXPERIENCE

JPMorganChase

Jun 2025 - Present | New Jersey, NJ

Quantitative Analytics Associate

- Architected and productionized a GPT-5 (**AzureOpenAI**) + Ensemble **RAG** system (**LangChain**, Chroma Vector DB, BM25, Parent Document Retriever) to auto-generate structured risk review drafts from 80-400 page model documents, reducing enterprise model review cycle time by **33%** (3 months to 2 months) across **500+** analysts.
- Implemented a **TabTransformer** in PyTorch on 300K loan records (11.5:1 imbalance), benchmarking against **Logistic Regression** (0.58 ROC-AUC) and **XGBoost** (0.65), achieving 0.61 ROC-AUC while evaluating attention-based feature interaction learning for credit risk modeling
- Conducted assessment of a **Regression-based** Auto loans recovery forecasting model (20K records) supporting an \$85B auto-loans portfolio, executing marginal vs conditional **R²** analysis, **KS/CSI** stability tests, **VIF** multicollinearity checks, **heteroskedasticity** diagnostics, and backtesting to surface structural risks prior to CCAR-aligned regulatory review
- Reconstructed and audited an **XGBoost** regression pipeline (20K datapoints) predicting 5 venture finance metrics using log transforms, k-fold CV, Random Search hyperparameter tuning, Recursive Feature Elimination, and **SHAP** explainability, identifying 7 risks before publication
- Engineered a concurrent API-driven reporting service (**Pandas**, **Multithreading**, OpenPyXL) that automated 7 daily financial exports into a unified multi-sheet workbook, cutting turnaround time from 45 minutes to <1 minute (**98% efficiency gain**) with retry logic and request deduplication

JPMorganChase

Jun 2024 - Aug 2024 | New Jersey, NJ

Quantitative Analyst Intern

- Built and productionized an LLM-driven Named Entity Recognition benchmarking system (**GPT-4 AzureOpenAI**, PyMuPDF) to evaluate 8 data extraction models on structured document parsing tasks, achieving 94% F1 on **CoNLL-2003** and establishing a standardized evaluation framework deployed via internal hosting for cross-model performance comparison
- Developed a computer vision-based stress testing engine (**OpenCV**, PyMuPDF) that generated 10+ noisy document variations per input to evaluate robustness of **8 data extraction models** and quantify performance degradation under real-world noise conditions.

Siemens Digital Industries Software

Jul 2021 - Jul 2023 | Pune, India

Associate Software Engineer

- Designed a two-stage AI pipeline (**OpenCV** + **Tesseract/Google Vision OCR** + **SpaCy/BERT NER**) to convert scanned PDF manufacturing drawings into SolidEdge-native 2D CAD files by extracting geometric primitives (73% shape accuracy) and metadata tables (90% NER accuracy), integrating outputs via SolidEdge APIs to prototype a **top customer-requested automation feature** in the CAD modeling domain
- Developed an early **GPT-3.5-powered conversational assistant** integrated into a web-based CAD editor (Zel-X) that leveraged help documentation to guide users through complex modeling workflows, prototyping an LLM-driven support interface to improve user experience
- Delivered **9 new features** across **7 repositories** for Zel-X and Xcelerator Share, two cloud-based CAD SaaS platforms built with **VueJS**, **ExpressJS**, and Siemens Web Framework, while resolving **50+ high-severity issues** in collaboration with offshore teams and increasing automated test coverage to **98%** using **Jest** within an **Agile SCRUM** environment

BizAmica Software Pvt. Ltd.

Aug 2020 - Mar 2021 | Pune, India

Machine Learning Intern

- Engineered and productionized an **OCR + NER pipeline** (Google Vision API, Tesseract, SpaCy/BERT, Regex) to extract named entities with bounding boxes from 20-50 page scanned legal PDFs (150 documents), achieving **75% F1** and integrating into the flagship software to accelerate structured information extraction for insurance litigation workflows

Center for Development of Advanced Computing

Feb 2020 - May 2020 | Pune, India

Research and Development Intern

- Evaluated multiple **face embedding architectures** on the **VGGFace2** dataset to benchmark clustering separability (**t-SNE** visualization) and inference latency for large-scale face recognition, conducting embedding distribution analysis and runtime profiling to select **FaceNet** as the production candidate for in-house identity matching systems
- Benchmarked **20 similarity search algorithms** on VGGFace2 face embeddings to evaluate retrieval accuracy and inference latency, conducting comparative performance profiling and selecting **Spotify Annoy** as the optimal **ANN index** for scalable in-house face recognition system

SKILLS

Python, PyTorch, AzureOpenAI, LangChain, Chroma Vector DB, BM25, Parent Document Retriever, Retrieval-Augmented Generation, SpaCy, Regex, XGBoost, Statsmodels, TabTransformer, Logistic Regression, SHAP, KS Test, CSI Test, Backtesting, OCR, Google Vision API, Tesseract, OpenCV, PyMuPDF, Similarity Search, Approximate Nearest Neighbors (ANN), t-SNE, Pandas, OpenPyXL, AsyncIO, ThreadPoolExecutor

EDUCATION

University of Southern California

Aug 2023 - May 2025 | Los Angeles, USA

Master of Science in Computer Science

GPA: 3.73 / 4

Pune Institute of Computer Technology

Aug 2017 - Apr 2021 | Pune, India

Bachelor of Engineering in Computer Engineering

GPA: 9.14 / 10